



School of
Information Technology
King Mongkut's Institute of Technology Ladkrabang

Machine Learning & Deep Learning for Natural Language Processing

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Topics

01. Machine Learning & Deep Learning

Supervised Learning

Logistic Regression

Neural Network

Deep Architecture

Unsupervised Learning

CBOW, Skip-gram

02

Training Prep

Data preprocessing, cleaning, and tokenization pipeline setup.

03

Model Training

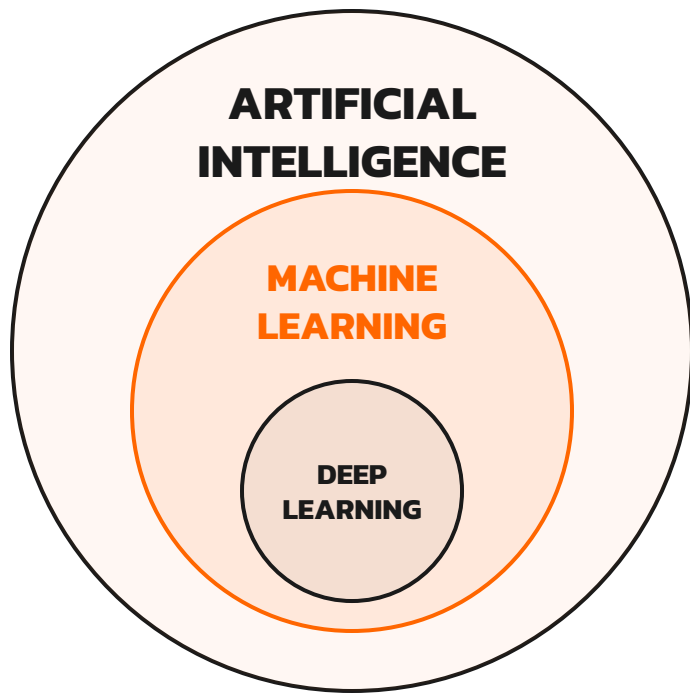
Executing optimization loops, hyperparameter tuning, and fitting.

04

Model Evaluation

Assessing accuracy, loss metrics, and final model validation.

Definition



CONCEPT 01

Machine Learning (ML)

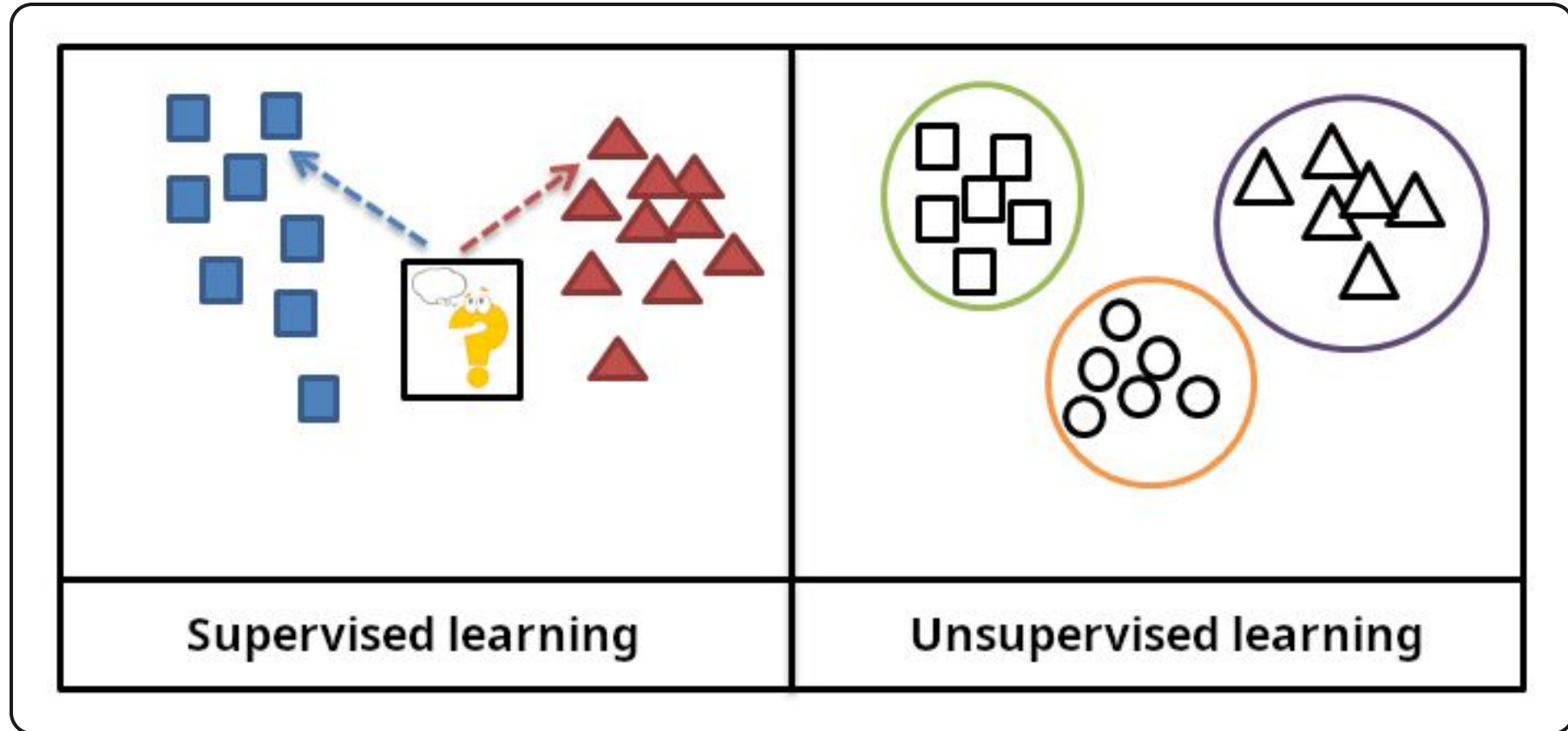
is the process of training a piece of software, called a model, to make useful predictions or generate content.

CONCEPT 02

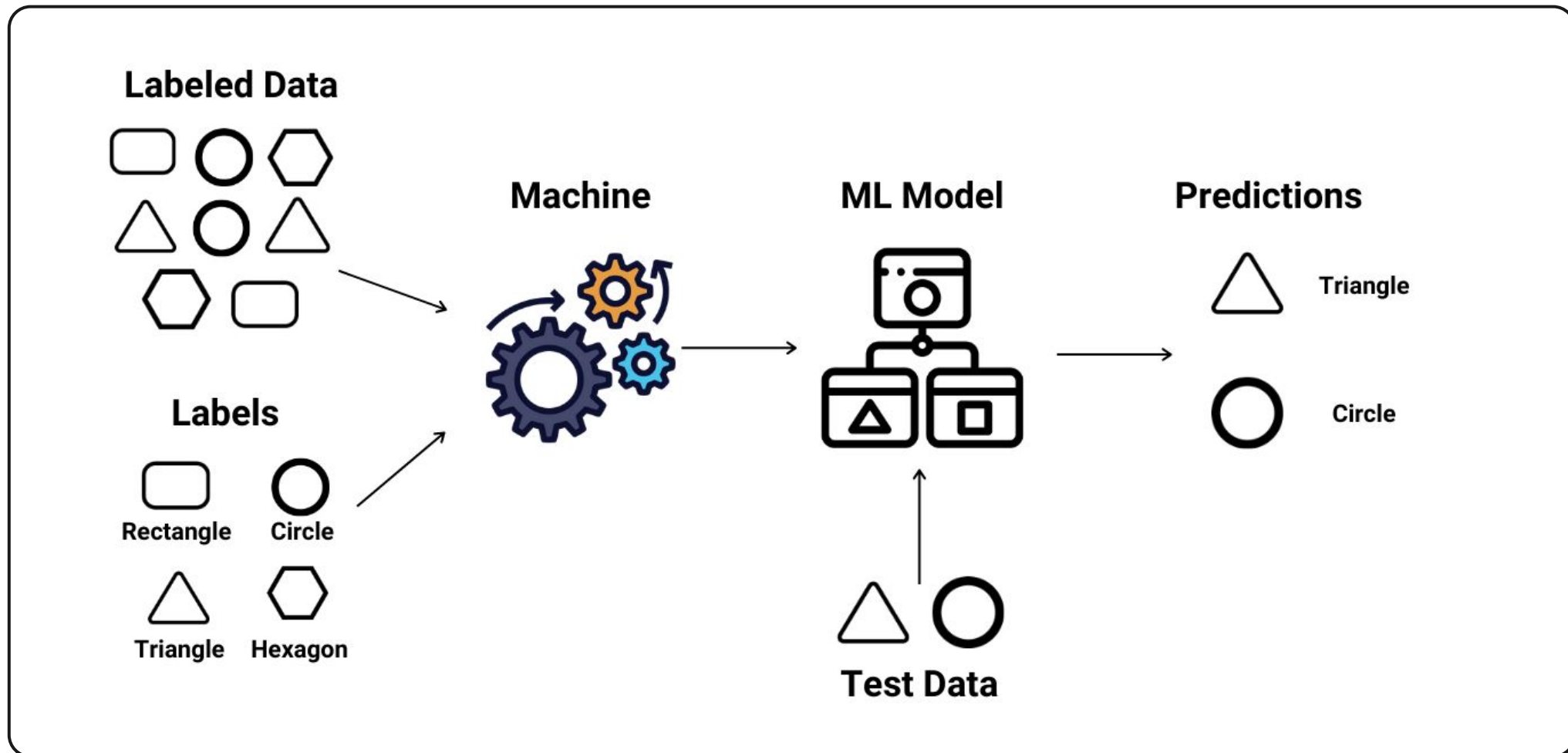
Deep Learning (DL)

is a specialized subset of ML that trains a model using multi-layered artificial neural networks to automatically learn complex patterns from raw data.

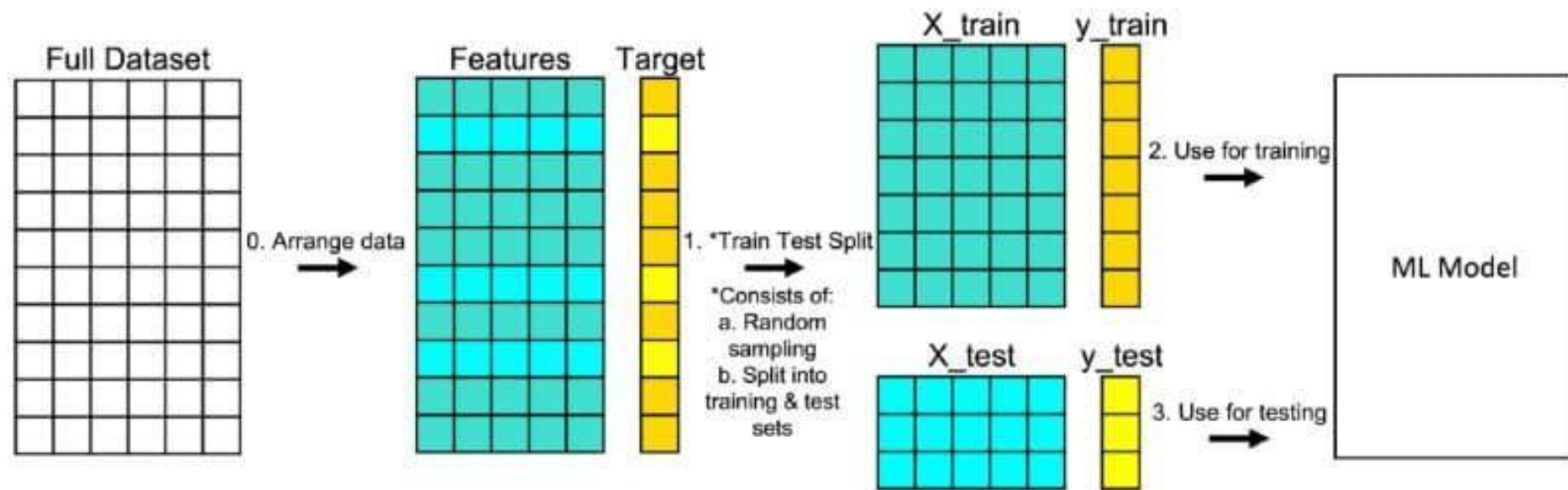
Machine Learning Type



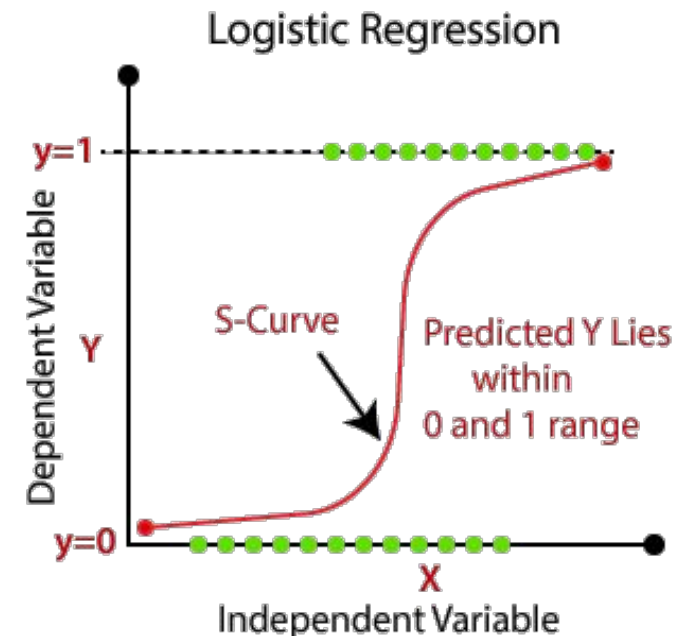
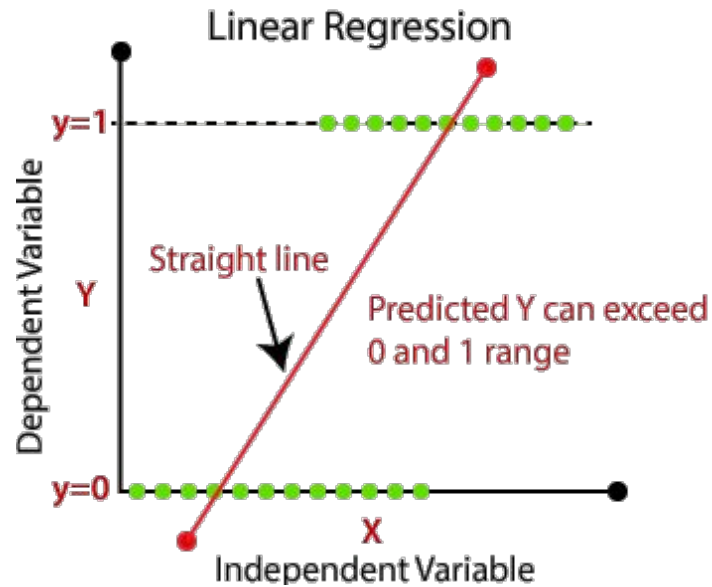
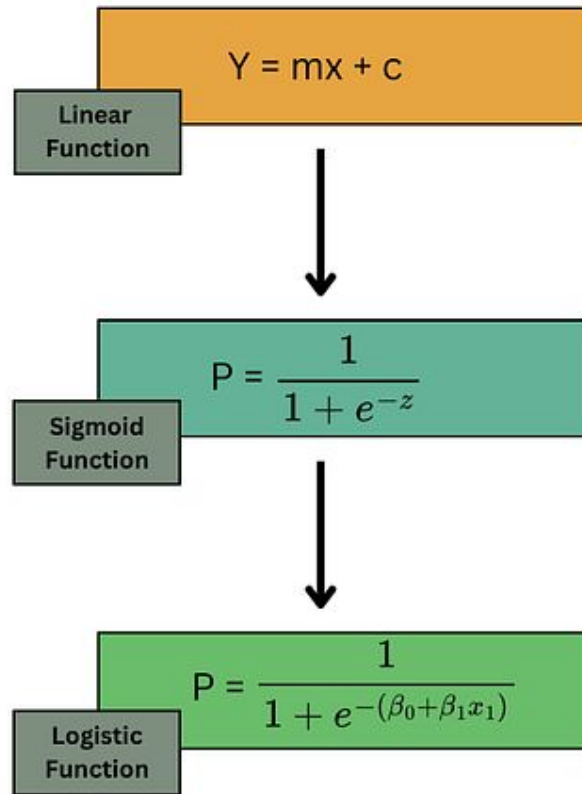
Supervised Learning Pipeline



Dataset for Supervised Learning



Model Algorithm

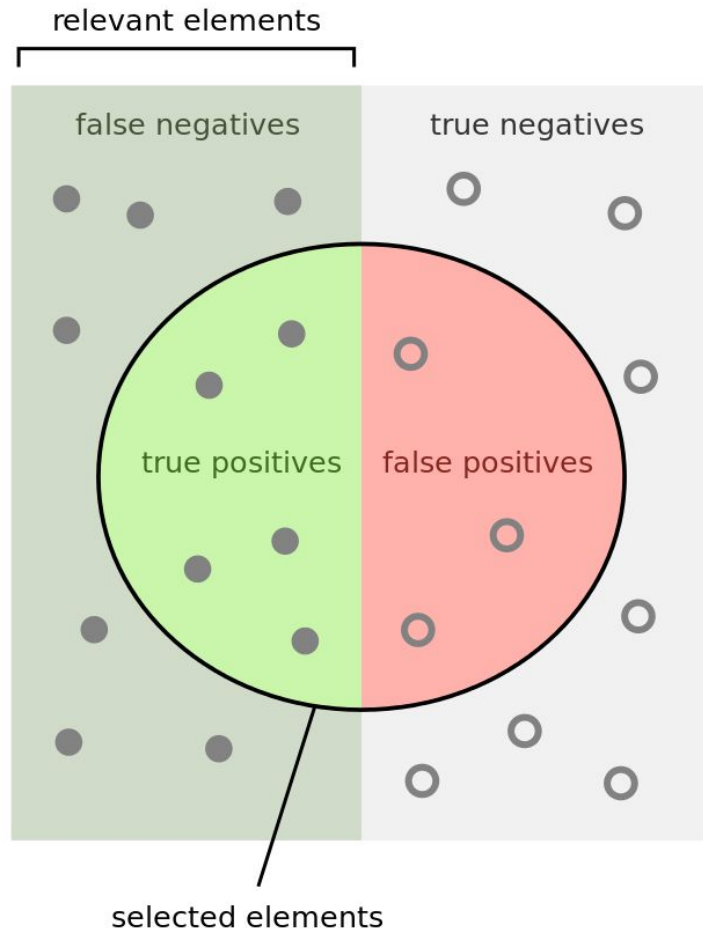


Model Evaluation

$$\text{Accuracy} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})}$$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Model Evaluation



How many selected items are relevant?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Model Evaluation Metrics

Accuracy (ความถูกต้อง)

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

Precision (ความแม่นยำ)

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (ความระลึก)

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score

ค่าเฉลี่ยแบบฮาร์มอนิก (Harmonic Mean) ระหว่าง Precision และ Recall เพื่อวัดประสิทธิภาพแบบสมดุล

$$\begin{aligned} \text{F1 Score} &= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} \\ &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

Workshop 1



Source Code

Access the complete Google Colab containing the implementation code, for sentiment analysis workshop.

<https://drive.google.com/file/d/1yQ82EDAmQkFLr3Aq9WgUteM8p-halVQk>

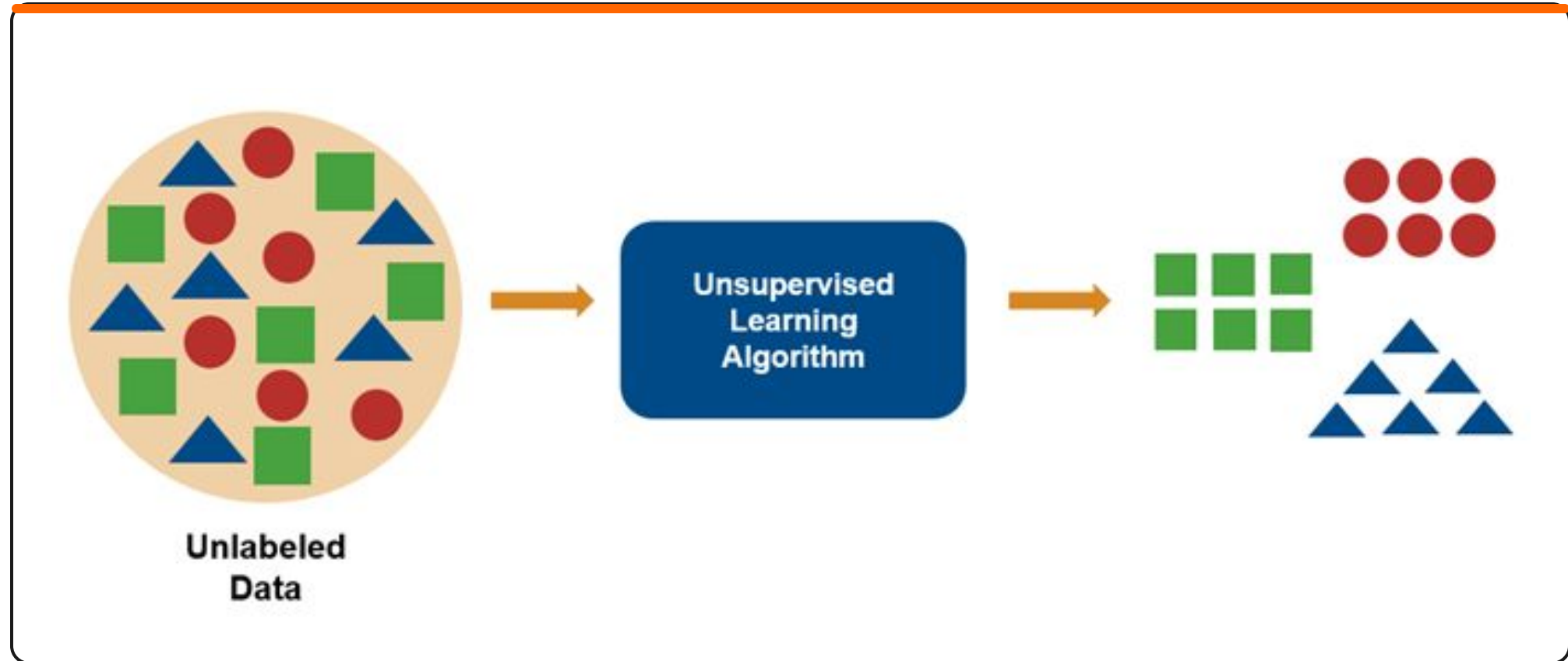


Dataset

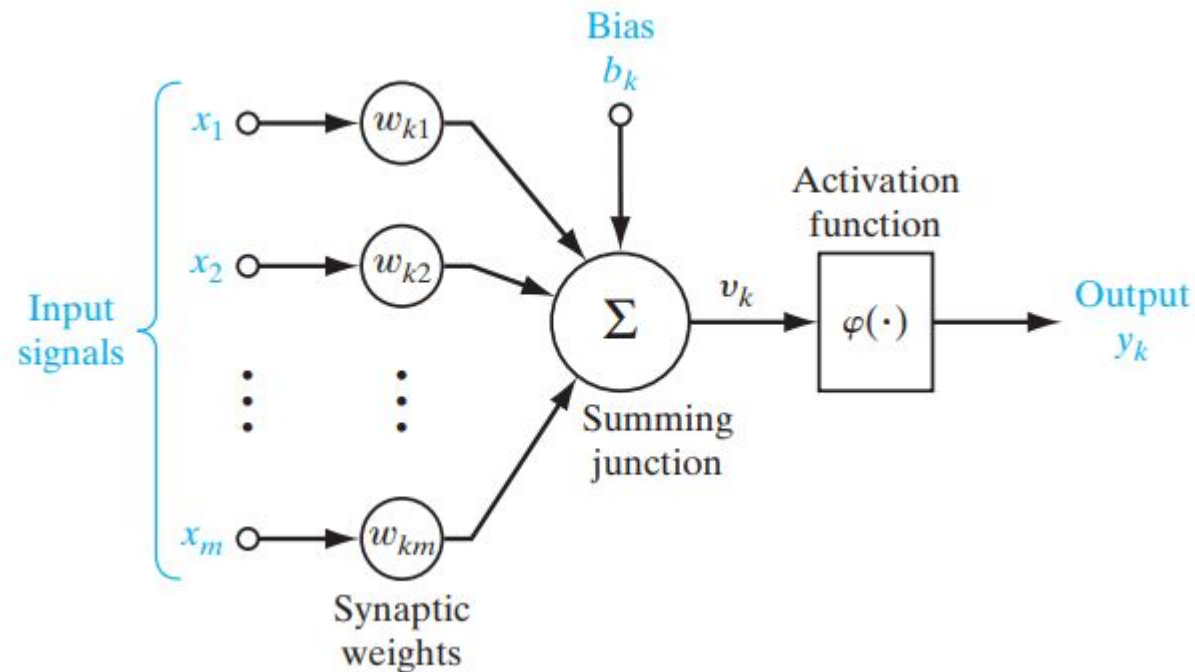
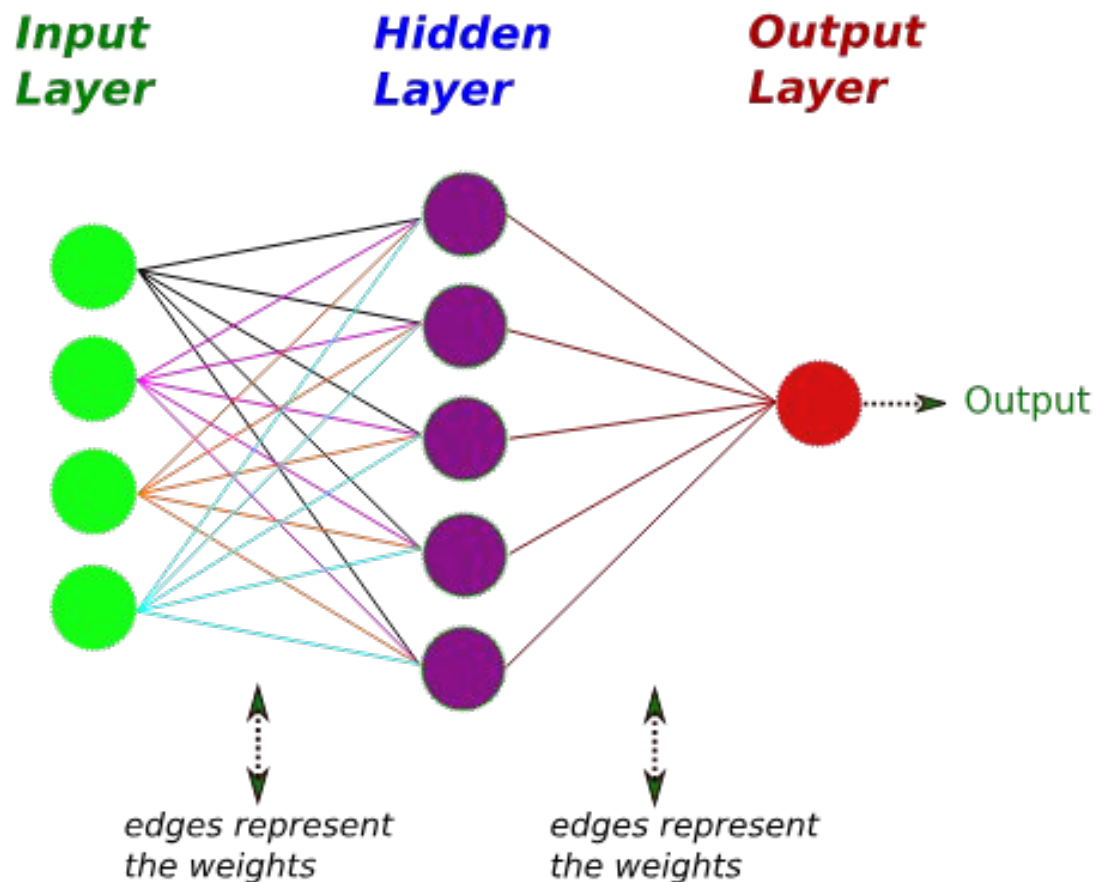
Access the Google Sheets dataset used for training, testing, and evaluating the sentiment analysis model.

https://docs.google.com/spreadsheets/d/1tRVy_DWX6Be6OA-JXMXGibTa1o73ZtPA

Unsupervised Learning Pipeline

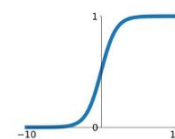


Neural Network



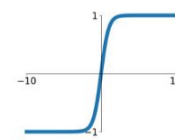
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



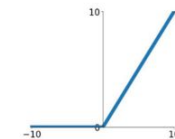
tanh

$$\tanh(x)$$



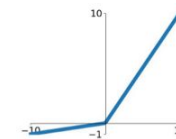
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

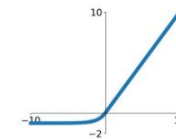


Maxout

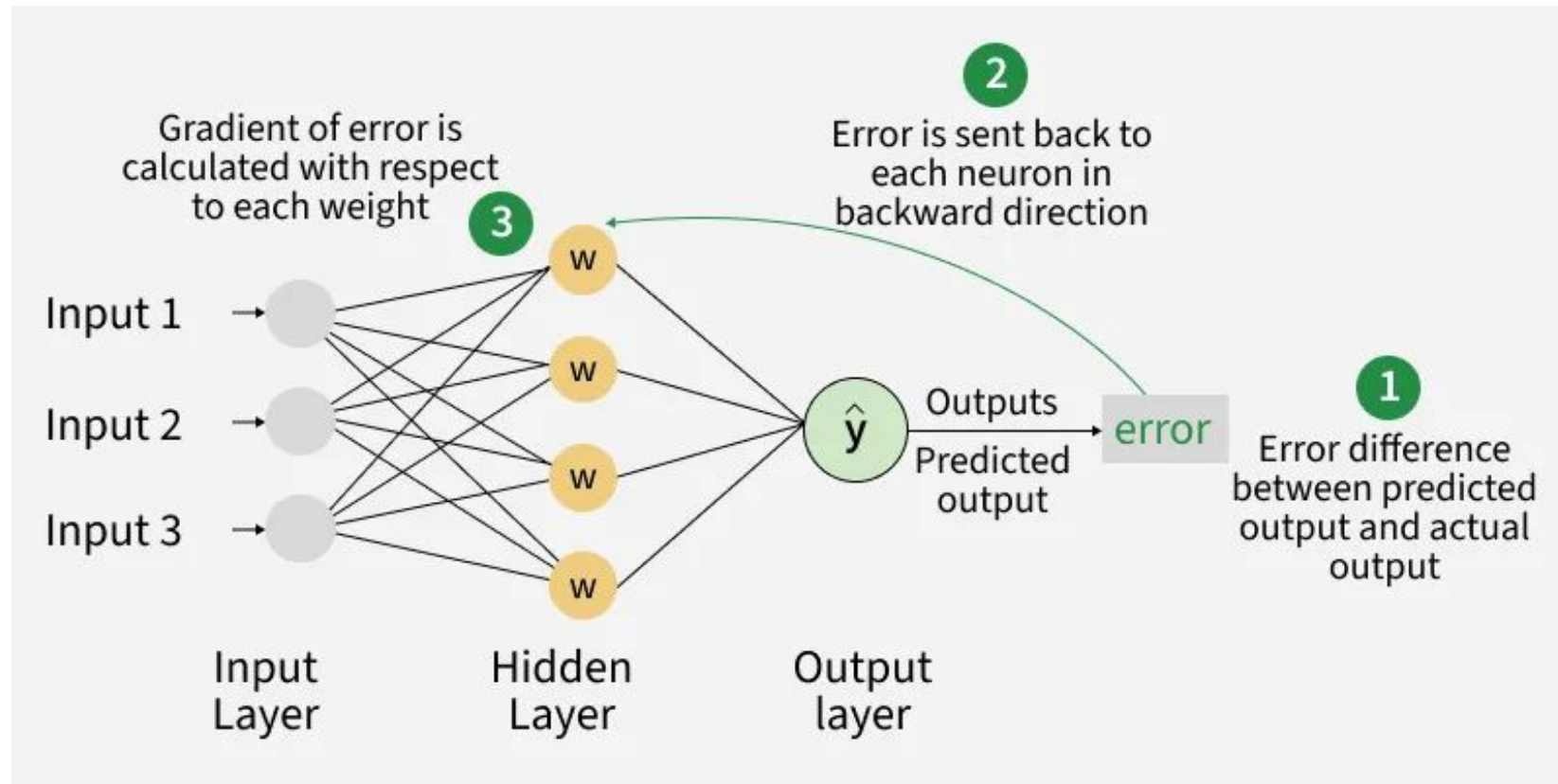
$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

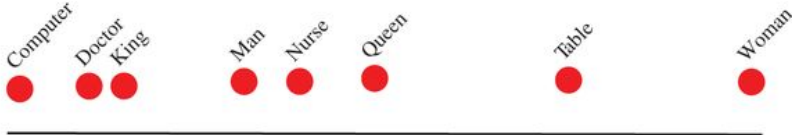


Backpropagation

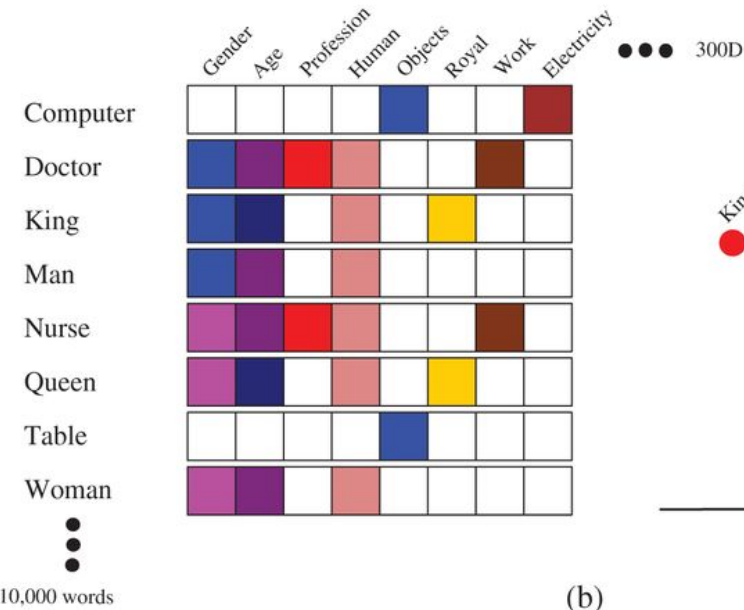


Embedding Model

Computer = 90
 Doctor = 450
 King = 580
 Man = 741
 Nurse = 982
 Queen = 1152
 Table = 1545
 Woman = 9854

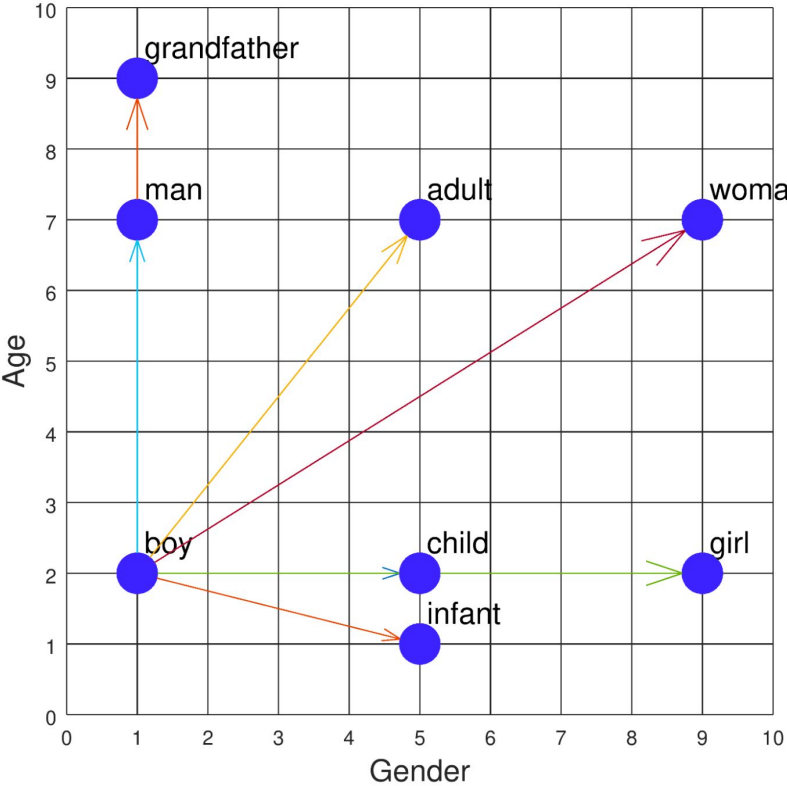


(a)



(b)

Vectors From 'boy'



Word2Vec

Vocabulary

Chihuahua

is

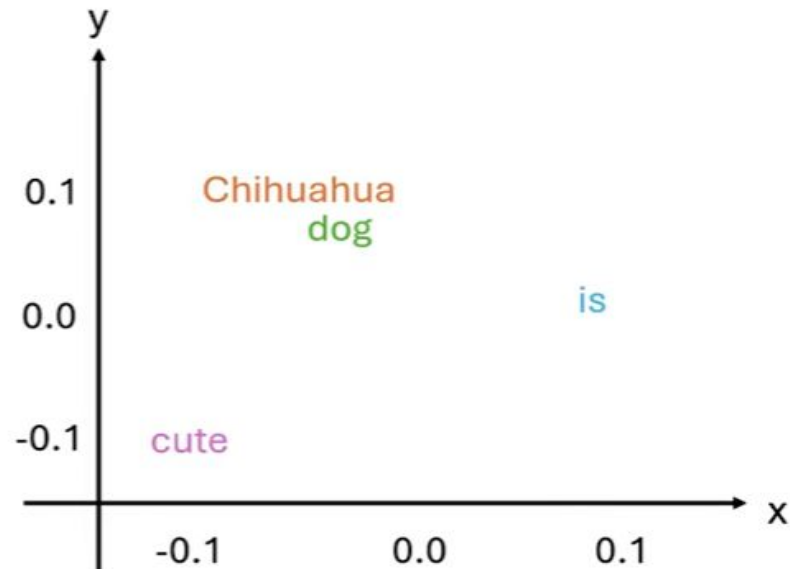
cute

dog

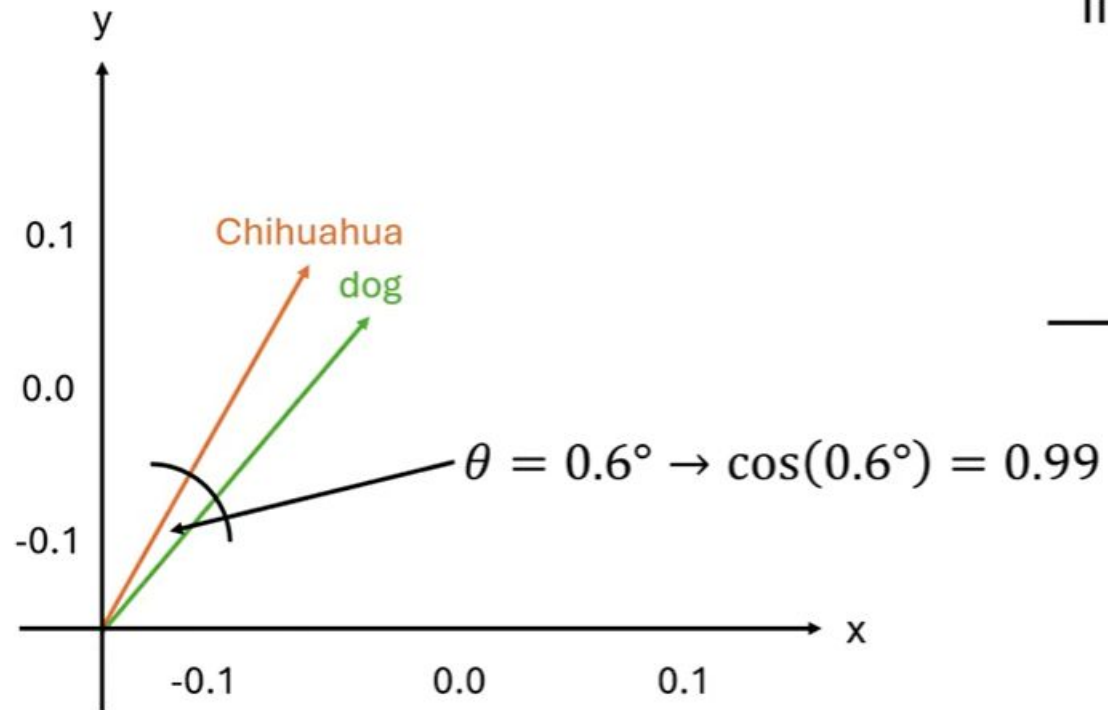


Vectors

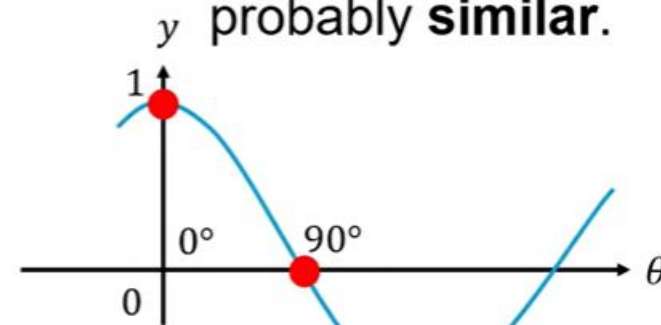
features	Chihuahua	is	cute	dog
	-0.08	0.1	-0.1	-0.09
	0.1	0.02	-0.1	0.09



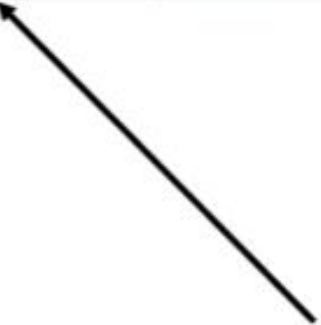
Apply Cosine Similarity



If the θ is **closer to 0°** , then it's probably **similar**.



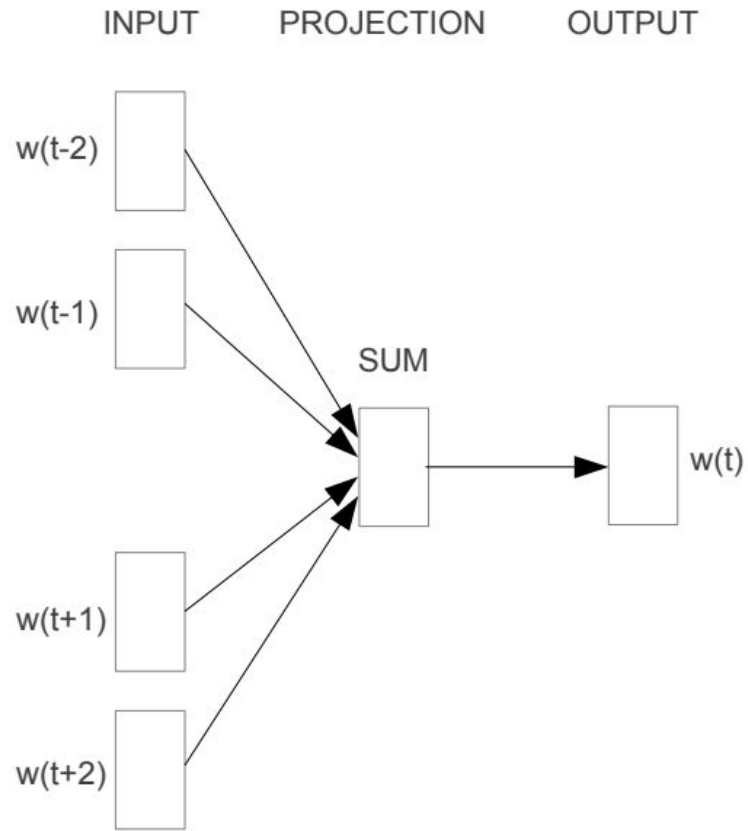
features	Chihuahua	is	cute	dog
x	-0.08	0.1	-0.1	-0.09
y	0.1	0.02	-0.1	0.09



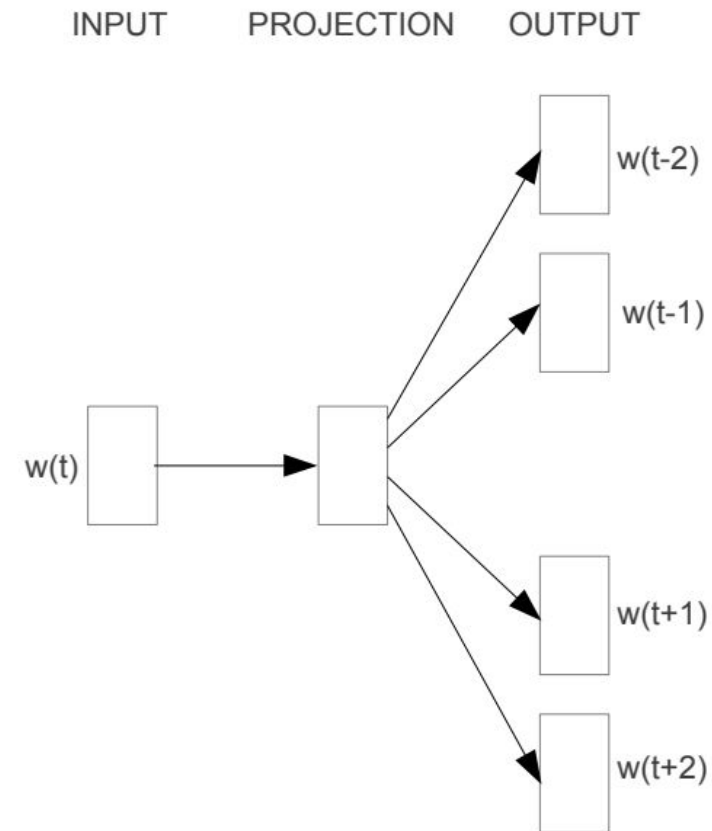
How to get all these value

Word Representations in Vector Space

<https://doi.org/10.48550/arXiv.1301.3781>

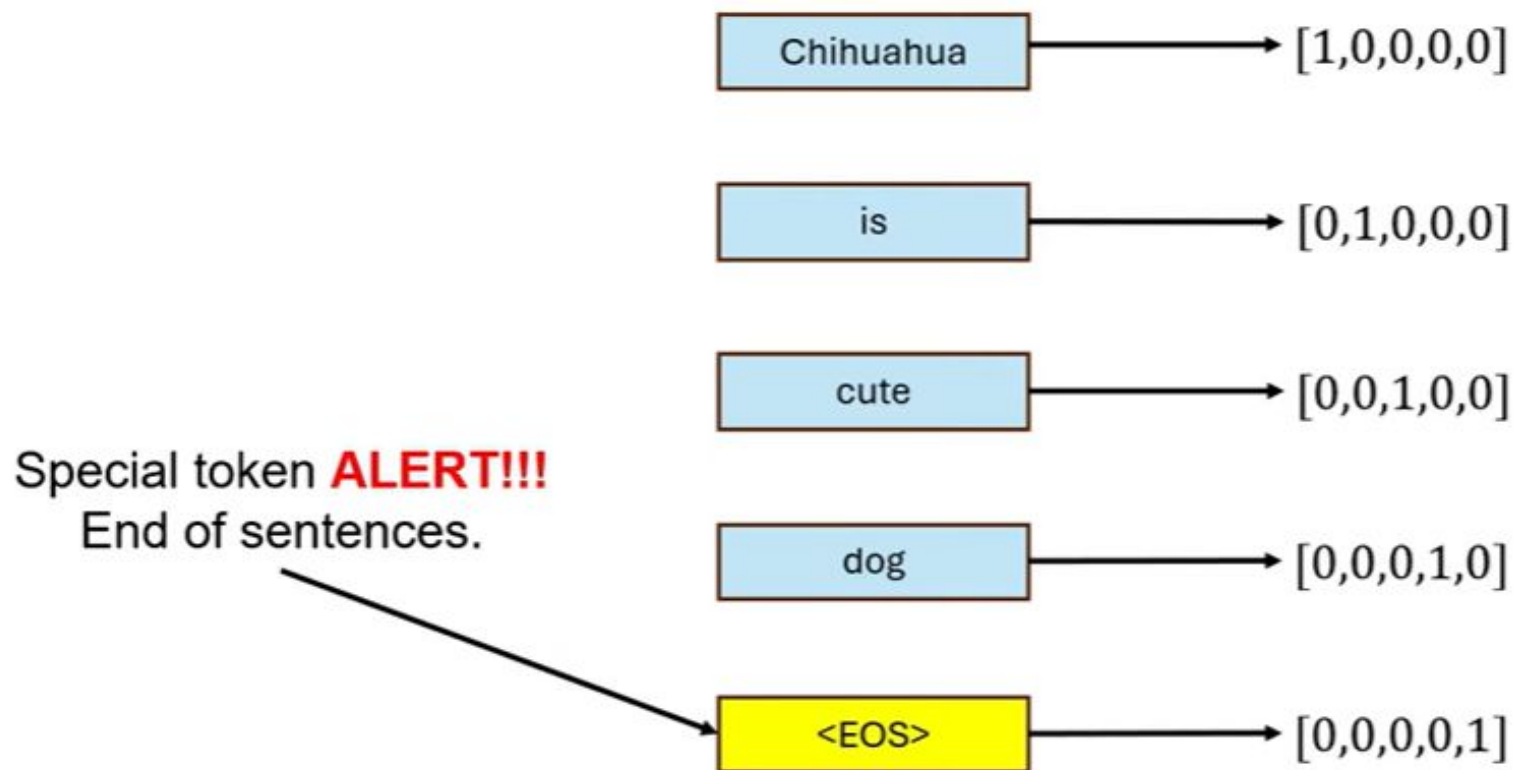


CBOW

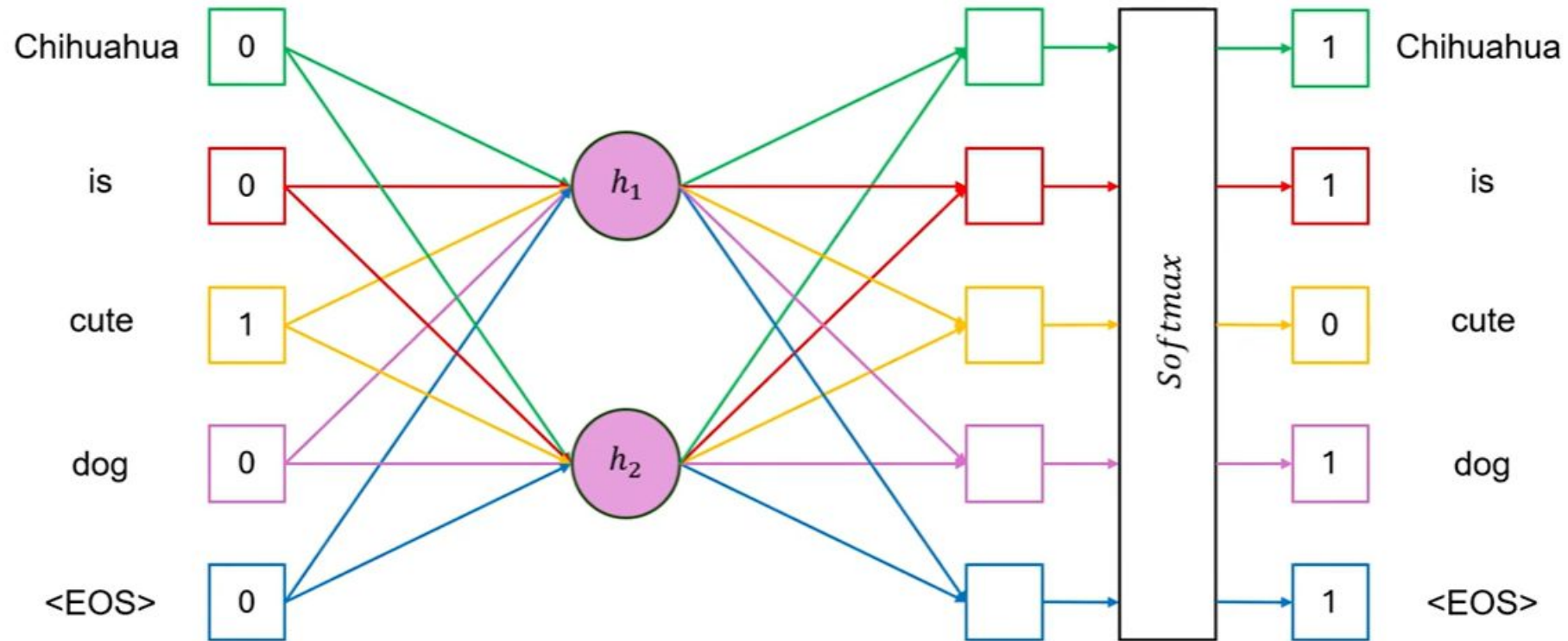


Skip-gram

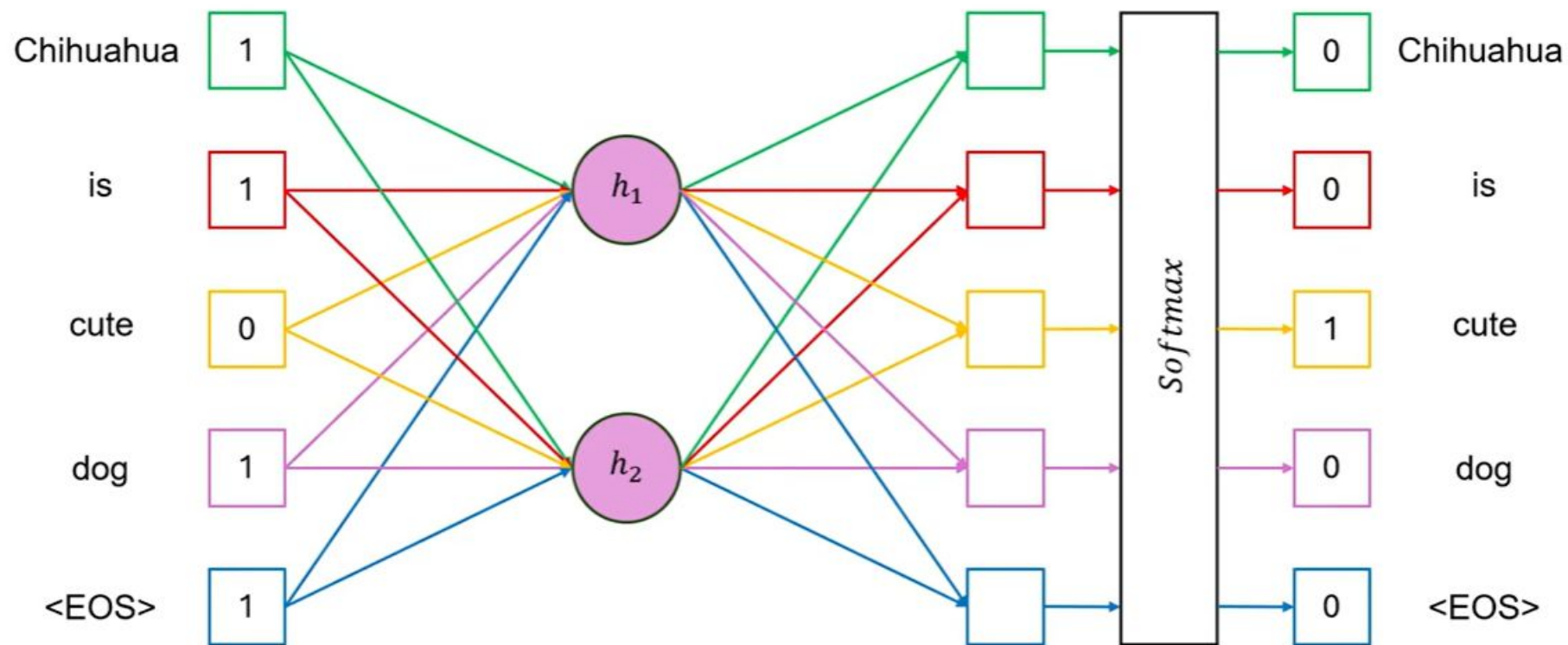
One-Hot encode!!!



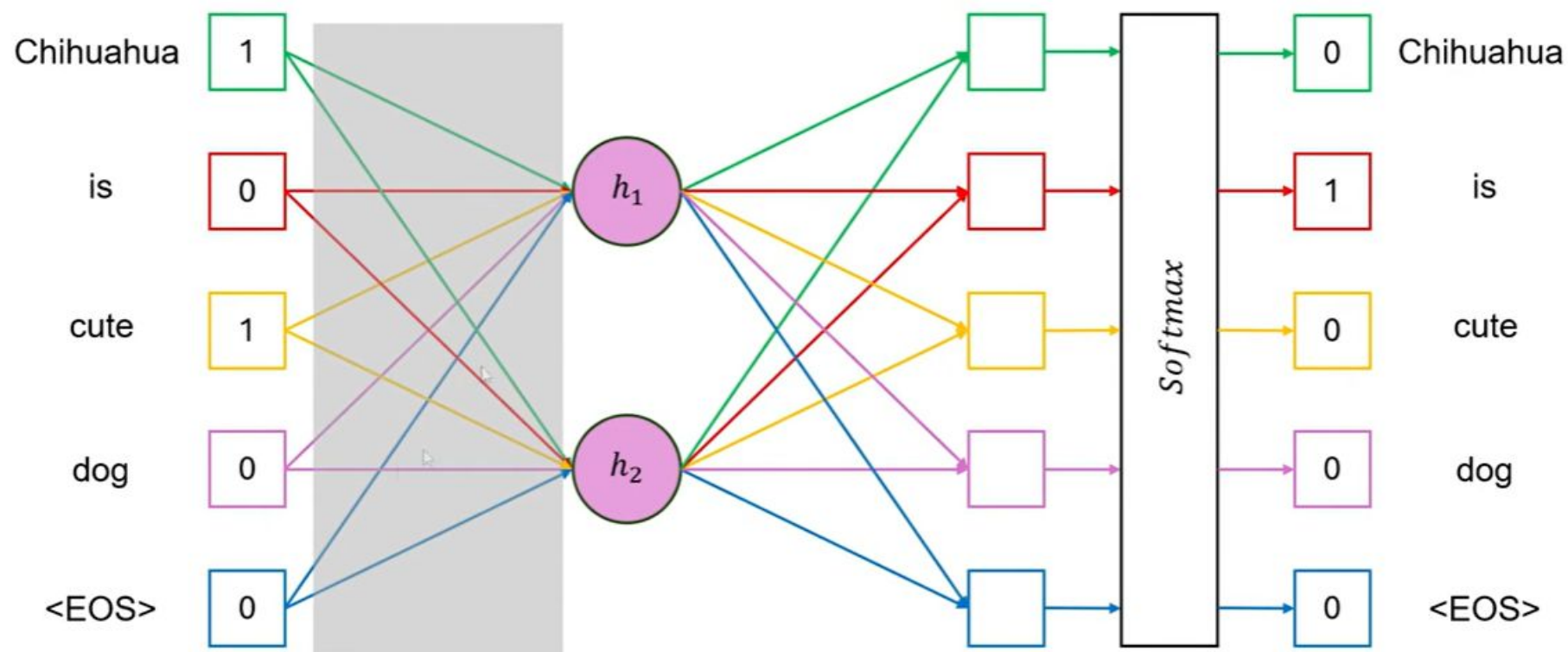
CBOW (Continuous bag of words)



SKIP-GRAM (Continuous skip grams)

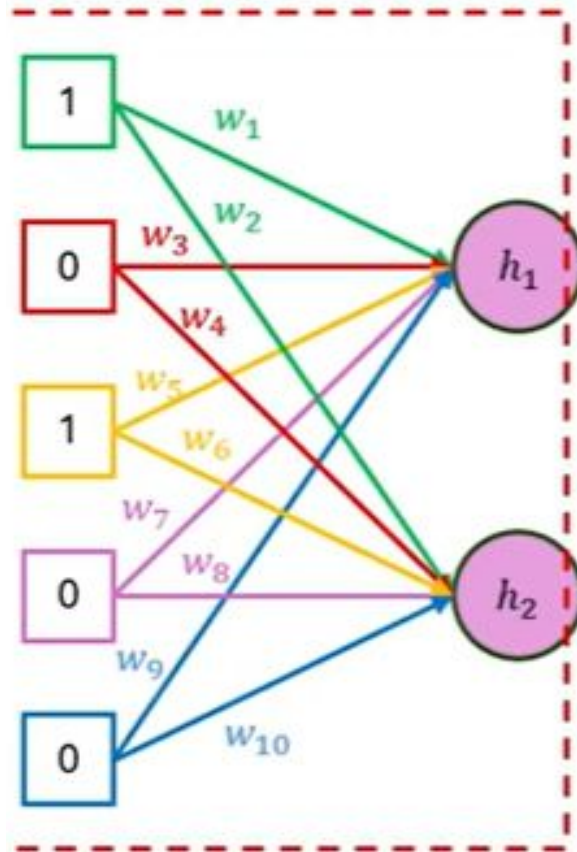


I'm going to use **CBOW**



Train and save the weight
of this state.

Weight Matrix



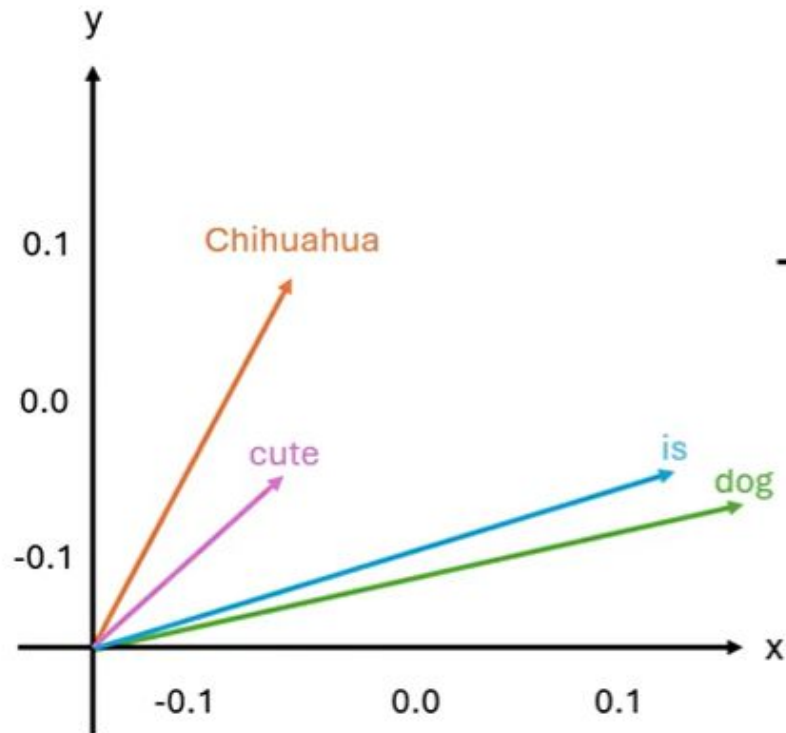
word (input)	→ h1	→ h2
Chihuahua	w1	w2
is	w3	w4
cute	w5	w6
dog	w7	w8
<EOS>	w9	w10

Word Embedding

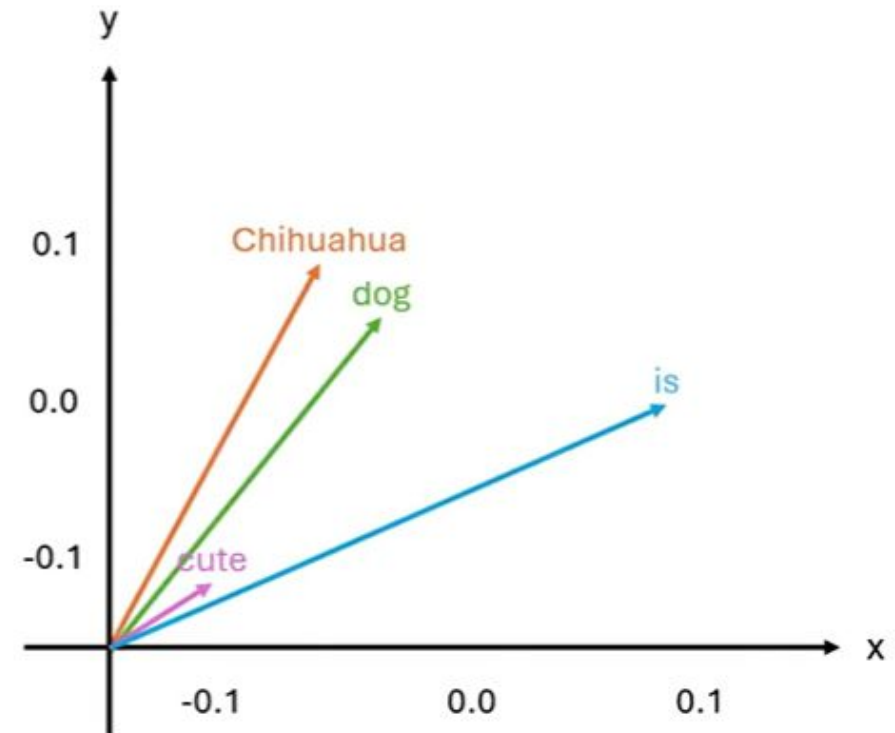
dimension	Chihuahua	is	cute	dog	EOS
h1	w1	w3	w5	w7	w9
h2	w2	w4	w6	w8	w10

Embedding Model

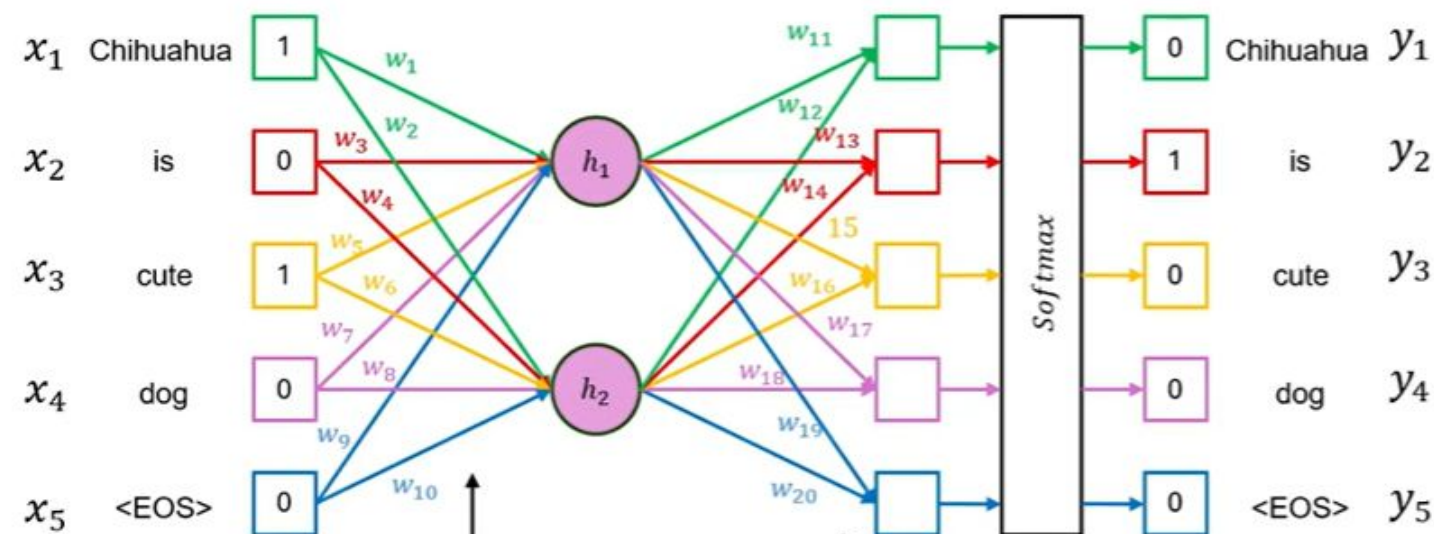
Before Training



After Training



CBOW Training Element



$$y_1 = \text{Softmax}(\text{sum}_1)$$

$$y_2 = \text{Softmax}(\text{sum}_2)$$

$$y_3 = \text{Softmax}(\text{sum}_3)$$

$$y_4 = \text{Softmax}(\text{sum}_4)$$

$$y_5 = \text{Softmax}(\text{sum}_5)$$

$$h_1 = (x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + x_5 w_9)$$

$$h_2 = (x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + x_5 w_{10})$$

$$\text{sum}_1 = (h_1 w_{11} + h_2 w_{12})$$

$$\text{sum}_2 = (h_1 w_{13} + h_2 w_{14})$$

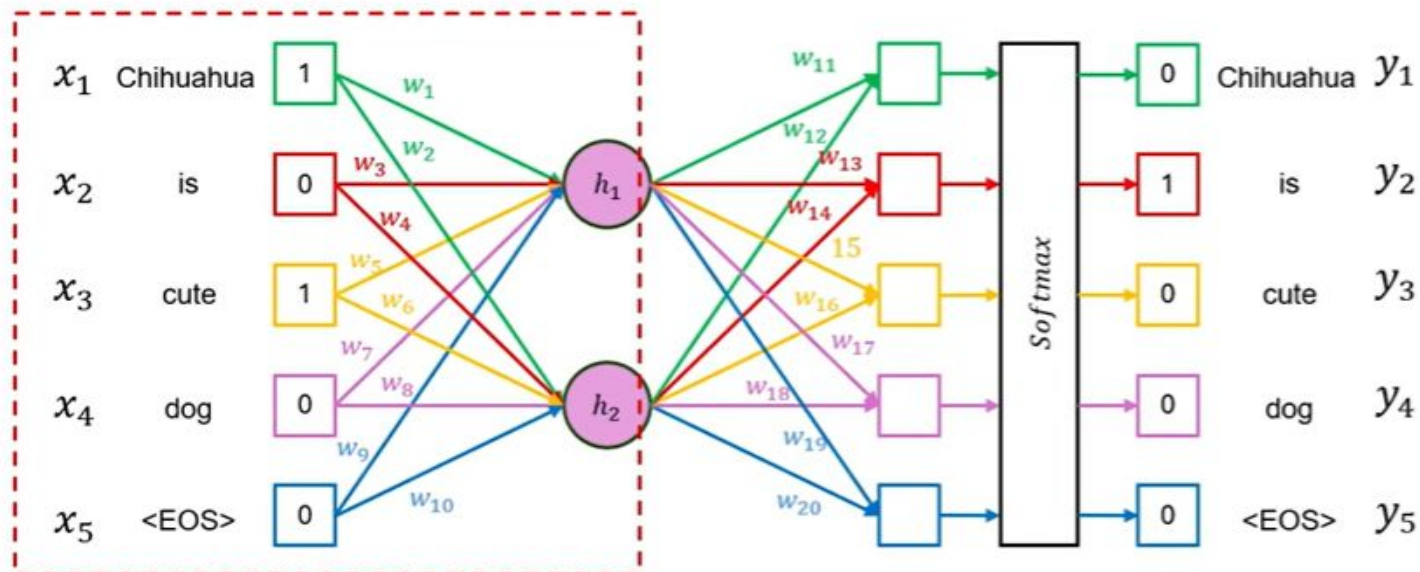
$$\text{sum}_3 = (h_1 w_{15} + h_2 w_{16})$$

$$\text{sum}_4 = (h_1 w_{17} + h_2 w_{18})$$

$$\text{sum}_5 = (h_1 w_{19} + h_2 w_{20})$$

CBOW Training Method

First



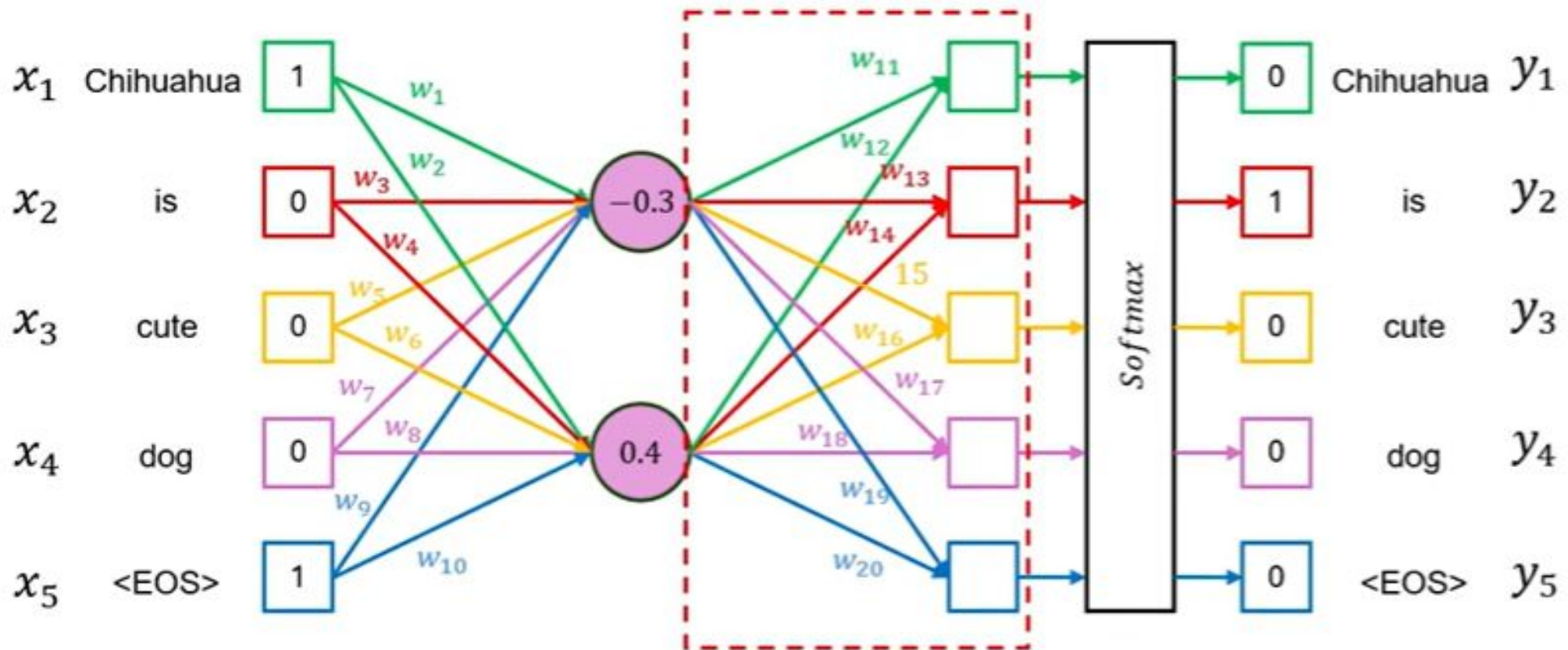
$$[x_1 \ x_2 \ x_3 \ x_4 \ x_5] \times \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \\ w_5 & w_6 \\ w_7 & w_8 \\ w_9 & w_{10} \end{bmatrix} = [h_1 \ h_2]$$

$$[h_1 \ h_2] = [1 \ 0 \ 1 \ 0 \ 0] \times \begin{bmatrix} 0.1 & 0.6 \\ 0.3 & -0.2 \\ -0.4 & -0.2 \\ -0.1 & 0.5 \\ 0.7 & -0.1 \end{bmatrix} = [-0.3 \ 0.4]$$

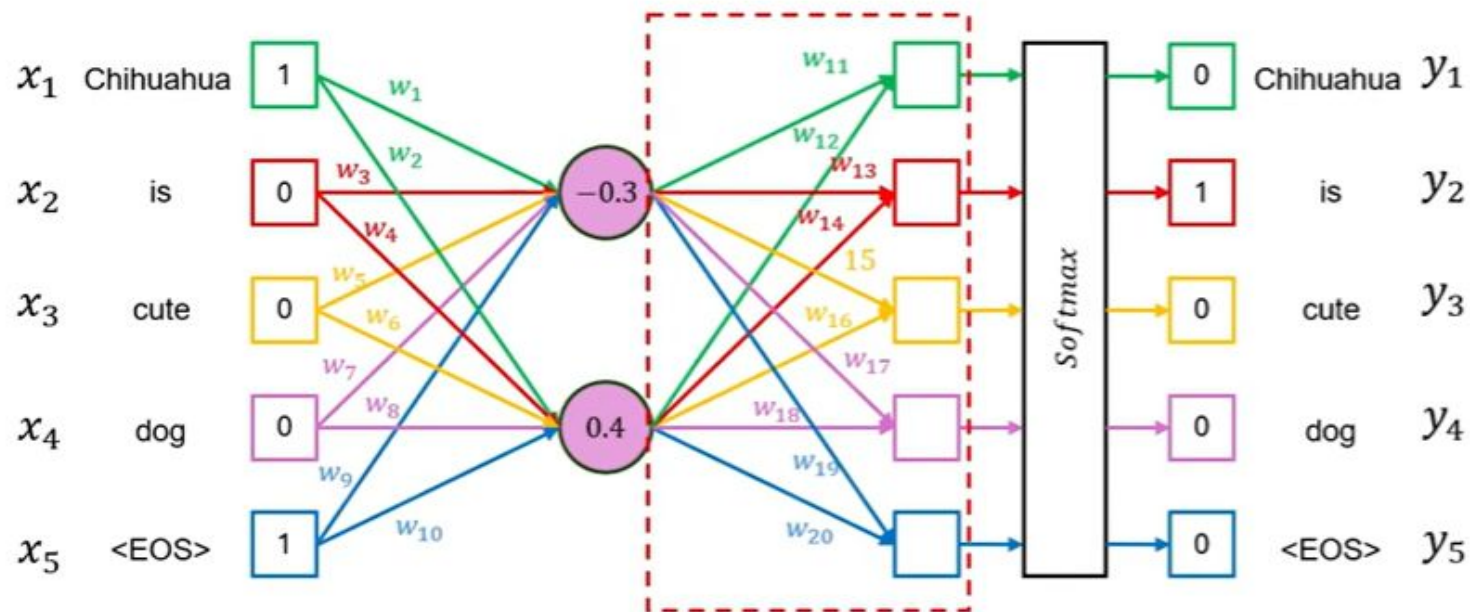
Random all weights

CBOW Training Method

Second



CBOW Training Method

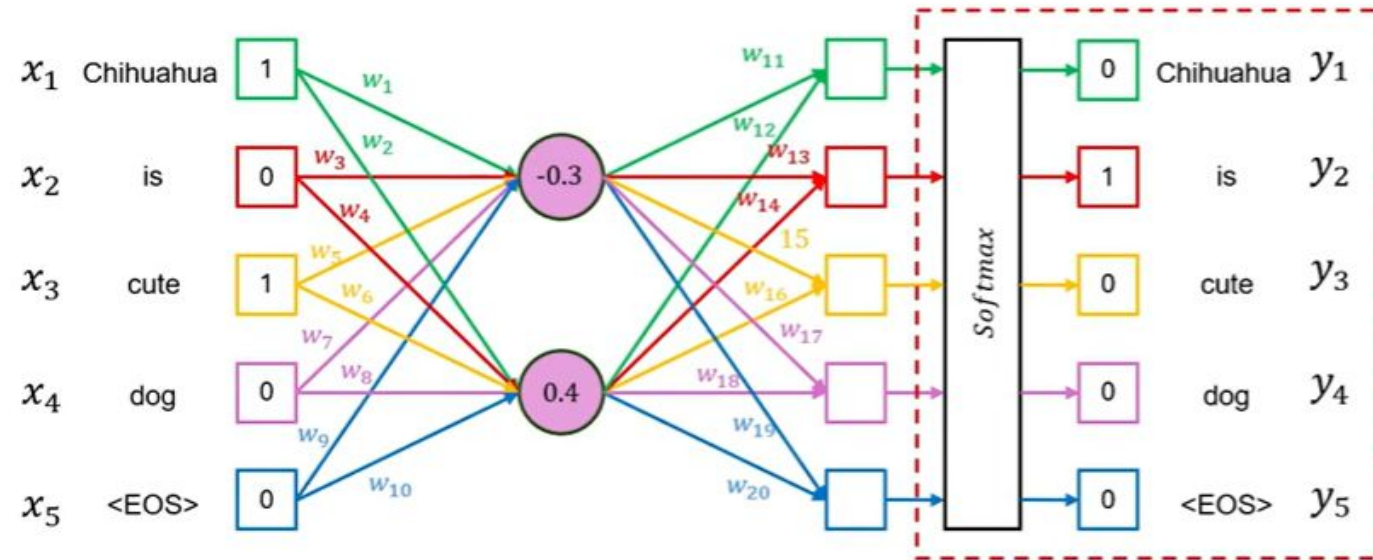


$$[sum_1 \quad sum_2 \quad sum_3 \quad sum_4 \quad sum_5] = [-0.3 \quad 0.4] \begin{bmatrix} w_{11} & w_{13} & w_{15} & w_{17} & w_{19} \\ w_{12} & w_{14} & w_{16} & w_{18} & w_{20} \end{bmatrix}$$

$$[sum_1 \quad sum_2 \quad sum_3 \quad sum_4 \quad sum_5] = [-0.3 \quad 0.4] \begin{bmatrix} 0.5 & 0.1 & 0.3 & -0.1 & 0.8 \\ 0.4 & -0.2 & -0.1 & 0.9 & 0.3 \end{bmatrix}$$

$$[sum_1 \quad sum_2 \quad sum_3 \quad sum_4 \quad sum_5] = [0.01 \quad -0.11 \quad -0.13 \quad 0.39 \quad -0.12]$$

CBOW Training Method



Third

$$[sum_1 \quad sum_2 \quad sum_3 \quad sum_4 \quad sum_5] = [0.01 \quad -0.11 \quad -0.13 \quad 0.39 \quad -0.12]$$

$$[S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5] = [0.20 \quad 0.17 \quad 0.17 \quad 0.29 \quad 0.17]$$

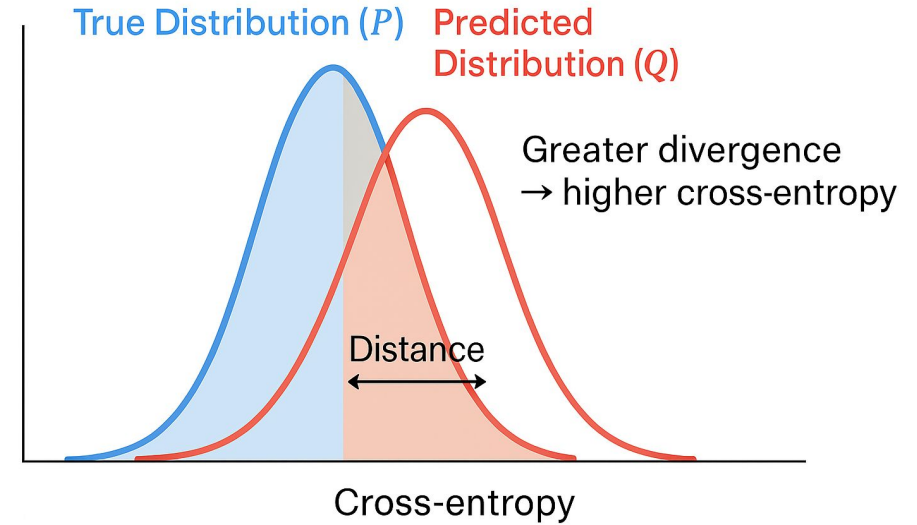
Then Feedback

$$[S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5] = [0.20 \quad 0.17 \quad 0.17 \quad 0.29 \quad 0.17] \xrightarrow{\text{Argmax}} [0 \quad 0 \quad 0 \quad 1 \quad 0] \xrightarrow{\text{Input}} (Chihuahua, cute) \rightarrow dog$$

But we need this answer: $[S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5] = [0 \quad 1 \quad 0 \quad 0 \quad 0] \rightarrow is$

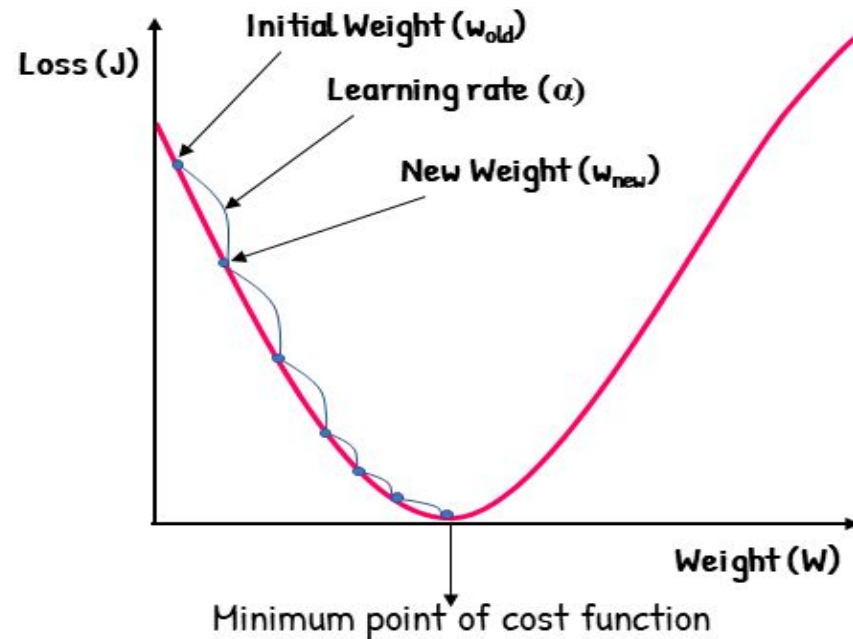
Loss Calculation

$$CE = - \sum_{i=1}^n \text{Observed} \cdot \log(\text{Softmax}_i)$$



Then, Backpropagation

$$w_{new} = w_{current} - \alpha \sum_{i=1}^n \frac{\partial CE_i}{\partial w}$$



Workshop 2



Source Code

Access the complete Google Colab containing the implementation code, for Word2Vec workshop.

<https://drive.google.com/file/d/1gkWcOndTgGS4z8sWRzaxsRjEgC83NHI->



Dataset

"the dog is cute" "the dog is loyal"
"the puppy is playful" "the puppy is cute"
"the cat is independent" "the cat is cute"
"the kitten is playful" "the kitten is cute"
"the lion is wild" "the lion is strong"
"the tiger is wild" "the tiger is strong"
"dogs and cats are popular pets"
"lions and tigers are wild animals"
"a puppy is a young dog"
"a kitten is a young cat"